

Assignment 12 Instructions: FINAL PROJECT DRAFT

Please answer the following questions in a Word document or a PDF and upload this to the Canvas site to submit this assignment by the deadline.

Project Overview

You will work through the complete data-project lifecycle on a real, multi-file open dataset of your choice (or select from the instructor's list). Starting with raw CSV/JSON files you will design a relational schema, load and clean the data, explore it with SQL, push the analysis further with Python, and finally reflect on ethical and performance considerations.

Tooling Requirements

- PostgreSQL (or SQLite if local install is a barrier)
- Python 3.x with: pandas, SQLAlchemy/psycopg2, matplotlib or seaborn, and one optional library of your choice (e.g., scikit-learn, plotly, pyspark).
- Jupyter Notebook or a reproducible .py script for each part.
- Git (private repo) for version control and submission.

Data Choices

- NYC 311 Service Requests (CSV, 8 M rows)
- COVID-19 U.S. county-level time series (CSV)
- LendingClub loan statistics (CSV + JSON)
- NOAA hourly weather ("ISD Lite") (fixed-width text)
- Reddit comment dumps (JSON)

You may propose a different public dataset if (a) ≥ 300 K rows or ≥ 1 GB uncompressed and (b) it contains at least three logical entity types. Approval due by Project Milestone 0.

Part 0 – Proposal & ER Diagram (10 pts, but required)

- 1-page proposal with a draft ERD and data source links.

Part 1 – Schema Design & Database Build (30 pts)

A. Conceptual $\rightarrow$ logical mapping (10 pts)

- Final ER diagram with entities, relationships, PKs, FKs.

B. SQL DDL script (15 pts)

- CREATE TABLE statements with constraints & appropriate data types.
- 2+ indexes and justification.

C. Data-volume sanity check (5 pts)

- Short Python script that counts rows & reports NULL/invalid counts per table.

Part 2 – Data Ingestion & Preparation (40 pts)

A. ETL in Python (25 pts)

- Read raw files, handle encoding issues, convert data types, standardize units/dates.
- Insert into DB in chunks (COPY or executemany) with timing logs.

B. Data-quality notebook (15 pts)

- Missing-value profile, outlier detection, at least two transformations (e.g., binning, scaling, text cleaning), written justification for each.

Part 3 – SQL Exploration & Analytics (40 pts)

Write and save eight SQL views or queries that demonstrate:

1. Single-table aggregation (GROUP BY)
2. Multi-table join (INNER)
3. Outer join or sub-query
4. Window/analytic function
5. CTE or recursive query
6. Derived metric updated with ALTER TABLE + UPDATE
7. Index-usage demonstration with EXPLAIN (before/after)
8. Creation of a materialized view or standard view + refresh strategy

Each query: 5 pts × 8 = 40 pts. Graded on correctness, readability, and brief written interpretation (2-3 sentences each).

Part 4 – Python Analysis & Visualization (40 pts)

A. Exploratory notebook (20 pts)

- Use pandas-SQLAlchemy connection to pull data.
- At least three visualizations (one must be interactive or geospatial if data allow).
- Narrative explaining trends, anomalies, and limitations.

B. Mini-model or advanced technique (20 pts)

Choose ONE: regression, classification, clustering, time-series forecasting, or network analysis. Include train/test split, metric evaluation, and a short explanation of why the method is appropriate.

Part 5 – Advanced Topics (30 pts)

Complete two of the following (15 pts each). Indicate your choices in the proposal.

1. NoSQL replication: push a subset of data to MongoDB, design an appropriate document model, and compare a sample query's performance vs. PostgreSQL.
2. Access-control & security: create two DB roles (analyst, guest) with different privileges and demonstrate them through failed/successful query screenshots; include short discussion of HIPAA/GLBA relevance to your dataset.
3. Efficiency study: benchmark three different indexing or partitioning strategies (e.g., hash vs. B-tree; range partitioning) and graph query latency.
4. SUGGEST your OWN ADVANCED TOPIC (It should be an implementation or a code based specific task that would qualify under advanced topics, you can propose this in our one to one final project meetings/reviews).